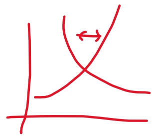
Ensemble

31 January 2026 17:31

Ensemble → Bagging → Bootstrap aggregation
Boosting →
Stacking

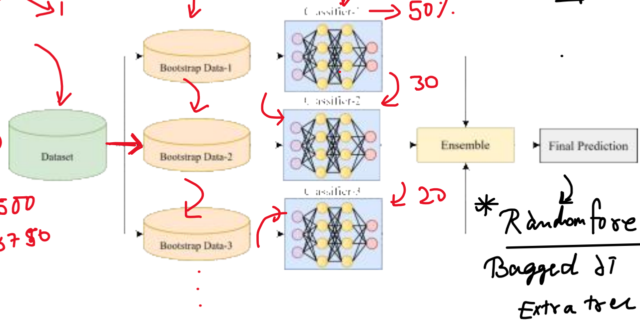


Bag

2 → variance

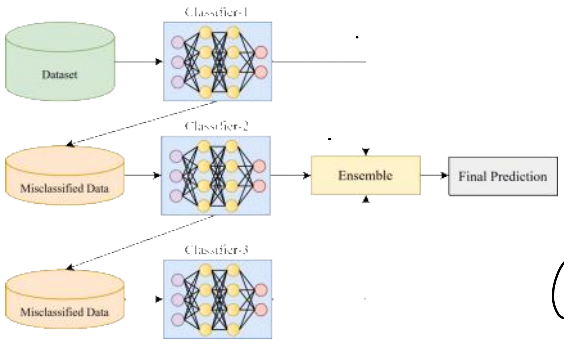
fraud → 0, 1

→ 70,000
1500
68780



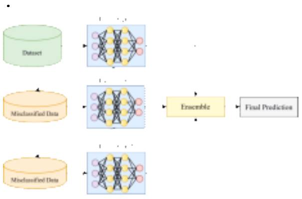
→ 50%
30
20

* Random forest ①
Bagged DT ②
Extra tree ③

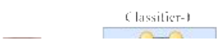


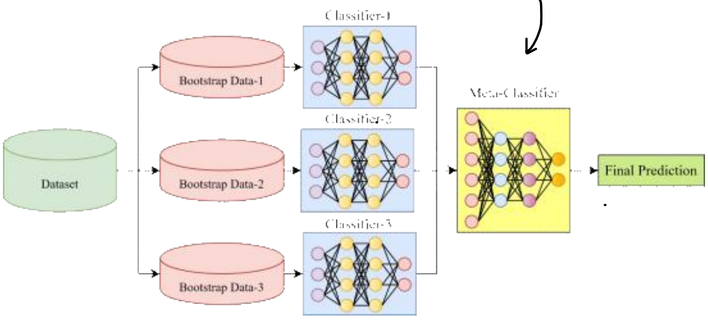
bias

1. Adaboost
2. SGD
3. GBM
4. XGBM
5. LGBM



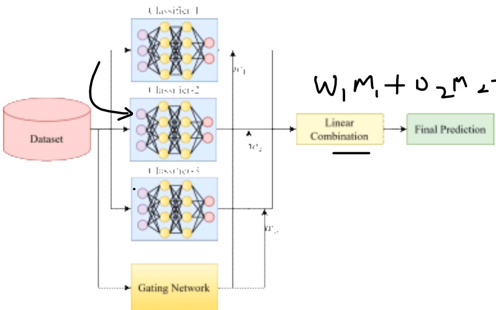
① → ②





4 MoE

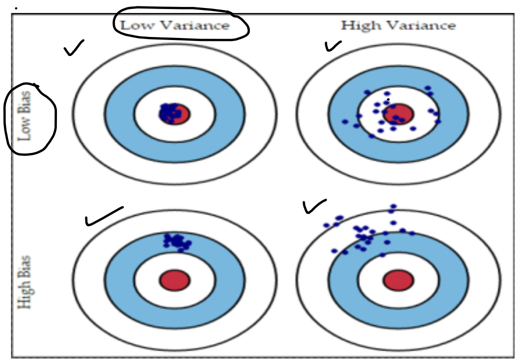
P → Revenue



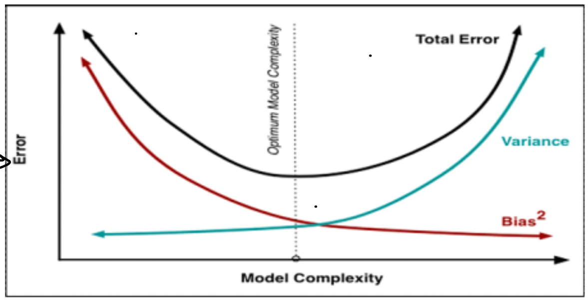
$W_1 M_1 + W_2 M_2 + W_3 M_3$

5. majority vote

$$Err(x) = \underbrace{E[\hat{f}(x) - f(x)]^2}_{\text{Bias}} + \underbrace{E[\hat{f}(x) - E[\hat{f}(x)]]^2}_{\text{Variance}} + \sigma_e^2$$



val
bias
Ve

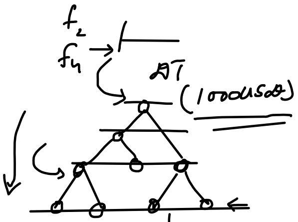


TT

→ No of Layer, parameters, S(θ), deg

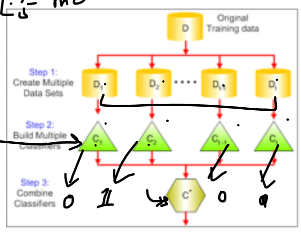
f_1
f_2
f_4 →

DT (1000users)



1 assumption
2 Telea
3 Google
4 FB
5 MC

Good
silver)



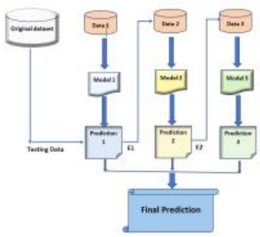
0 1 0 0 → 0

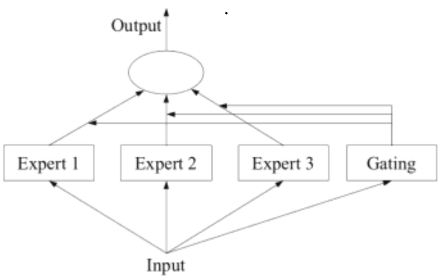
Bagg

① Sample with Sub
② oob' Reduction

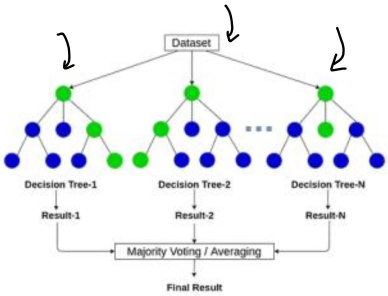
oob

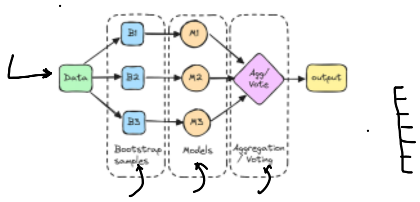
①
②
③ contr





Random Forest





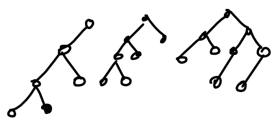
D_{trees}

$$Gain(T, x) = E(T) - E(T, x)$$

Impurity	Task	Formula	Description
Gini impurity	Classification	$\sum_{i=1}^{C} f_i(1 - f_i)$	f_i is the frequency of label i at a node and C is the number of unique labels.
Entropy	Classification	$\sum_{i=1}^{C} -f_i \log(f_i)$	f_i is the frequency of label i at a node and C is the number of unique labels.
Variance / Mean Square Error (MSE)	Regression	$\frac{1}{N}\sum_{i=1}^{N}(y_i - \mu)^2$	y_i is label for an instance, N is the number of instances and μ is the mean given by $\frac{1}{N}\sum_{i=1}^{N} y_i$
Variance / Mean Absolute Error (MAE) (Scikit-learn only)	Regression	$\frac{1}{N}\sum_{i=1}^{N}\lvert y_i - \mu\rvert$	y_i is label for an instance, N is the number of instances and μ is the mean given by $\frac{1}{N}\sum_{i=1}^{N} y_i$

SKLEAN → RF → clas

RFr → REG



$$n_{ij} = w_j C_j - w_{left(j)} C_{left(j)} - w_{right(j)} C_{right(j)}$$

$$\hookrightarrow f_{ij} = \frac{\sum_{j:node} f_{i(n_{ij})}}{\sum_{all} n_{ik}}$$